

Response Letter

Jul 13, 2026

Dear Wenjian Yu, PhD
Tsinghua University, Beijing, China
Editor-in-Chief
Integration

Ms. Ref. No.: **VLSIJ-D-26-00858**

Title: A Hybrid Hyperchaotic Pseudorandom Number Generator with Dynamic
Perturbation

Dear *Wenjian Yu, Dr.*,

By means of this letter, I am sending you the revised manuscript with Ref. No. VLSIJ-D-26-00858 which has been submitted for possible publication as Research paper in the prestigious journal of *Integration*. In this new version of the manuscript, we have incorporated all the suggestions and valuable comments provided by the anonymous reviewers. In the following pages are our point-by-point responses to each of the comments of the reviewers.

The authors would like to thank the reviewers for their time and their valuable comments which highly insightful and enabled us to greatly improve the quality of our manuscript.

I thank you in advance for all the attentions you may give to this letter.

Yours sincerely,
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Response to comments of reviewers

The authors are grateful to the anonymous reviewers for their valuable comments and suggestions, which highly insightful and enable us to greatly the quality of our manuscript.

Responses to the comments of Reviewer #2:

Reviewer' comments:

The manuscript proposes a hybrid pseudorandom number generator that integrates 4D continuous and 2D discrete hyperchaotic systems. Before publication, some issues need to be addressed:

1. The contributions and novelties of the work should be further illustrated.

Response to this suggestion: *A contributions subsection has been added in 1.1 to better illustrate the contributions*

2. Explain why these two systems are chosen over other high-dimensional or discrete chaotic maps from the perspective of hardware resource consumption and dynamic performance

Response to this suggestion: *In section 3 a more detailed explanation for the choices of chaotic systems*

3. It is important to sufficiently introduce the existing related work to highlight the study topic, some recent work can be considered, A unified framework for generating 4D discrete memristive hyperchaotic maps with complex dynamics and application to encryption; Encryption design and analysis of 3D medical models in Internet of Medical Things using a novel memristive hyperchaotic map; A family of image encryption schemes based on hyperchaotic system and cellular automata neighborhood.

Response to this suggestion: *Recent related works have been added to Section 2 to illustrate current research*

4. Is the proposed can be effective to chaotic neural networks, please discuss this by related to the literature, Universal method for enhancing dynamics in neural networks via memristor and application in IoT-based robot navigation

Response to this suggestion: *Added future work in section 6 to better illustrate future applications and integrations, such as the one mentioned*

5. The selection reasons for the parameter values need to be explained.

Response to this suggestion: *Added section 3.4 to better explain the paramter values*

6. PRNG and other abbreviations should give their full name when it first appears in the text.

Response to this suggestion: *Added abbreviations of such concepts as PRNG and FPGA in the abstract*

Response to comments of reviewers

The authors are grateful to the anonymous reviewers for their valuable comments and suggestions, which highly insightful and enable us to greatly the quality of our manuscript.

Responses to the comments of Reviewer #3:

Reviewer' comments:

This paper presents a well-structured, hardware-aware hybrid PRNG combining a 4D continuous hyperchaotic system with a 2D discrete perturber—a novel dynamic coupling strategy that effectively mitigates digital degradation on FPGA.

1. The design of the fixed-point number format lacks the modeling of quantization error propagation, posing a risk of implicit chaotic degradation. It is recommended to supplement the fixed-point sensitivity analysis and the dynamic bit-width adaptive mechanism

Response to this suggestion: *The precision needed for the system is discussed in section 3.1. The dynamic bit-width adaptive mechanism is discussed in section 6*

2. The post-processing architecture has timing bottlenecks and potential side-channel leakage issues. It is recommended that the authors discuss this.

Response to this suggestion: *Timing analysis has been shown in the work and discussion on the side-channel attacks is discussed on section 6*

3. The security verification lacks key industrial standard tests. It is recommended to supplement the NIST SP 800-90B entropy assessment and the FPGA physical layer antiside-channel test.

Response to this suggestion: *Section 5.9, added to show test*

Response to comments of reviewers

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Responses to the comments of Reviewer #4:

Reviewer' comments:

This manuscript presents a hybrid pseudorandom number generator combining continuous and discrete hyperchaotic systems. The design employs dynamic parametric perturbations and modular multiplication to mitigate digital degradation. The generator is implemented on an FPGA platform. The authors validate randomness using standard statistical suites like NIST and TestU01. This manuscript presents an interesting hardware design of a hybrid hyperchaotic generator, showing promising potential for cryptographic applications. The proposed dynamic perturbation mechanism effectively addresses digital degradation. However, several critical technical details require substantial improvement before publication. Specifically, the numerical discretization methodology, hardware performance evaluation lacking operating frequency data, and statistical testing protocols need deeper theoretical justification and more rigorous experimental validation to prove practical feasibility. Therefore, my recommendation is Major Revision. I strongly encourage the authors to carefully address the following detailed concerns.

1. The abstract should be refined to concisely address the research background, identify the gap, state the motivation, outline the study's content, summarize key findings, and highlight the significance of the work. Please refer to Exploiting robust quadratic polynomial hyperchaotic map and pixel fusion strategy for efficient image encryption and Novel approach to time series forecasting based on memristive Hopfield Neural Network with hidden heterogeneous and homogeneous extreme multistability for improvements.

Response to this suggestion: *Abstract changed to better identify highlights and recommended references were added.*

2. I think the keyword list should be adjusted. Generally, we should select 5-7 precise keywords that accurately represent the study's central problem, technical approach, and principal results, thereby optimizing indexing, discoverability, and relevance to the journal's scope. Please refer to Lossless frequency-domain image encryption via 3D exponential hyper-chaotic map and integer lifting wavelet transform and Extreme multistability in discrete memristive neuron maps and implications for dual-field applications for revisions.

Response to this suggestion: *More precise keywords were used*

3. The introduction section lacks some essential elements. For example, the authors should provide a brief introduction, analysis, and discussion of chaotic system construction and its extensive applications, such as Exploiting robust quadratic polynomial hyperchaotic map and pixel fusion strategy for efficient image encryption, Novel approach to time series forecasting based on memristive Hopfield Neural Network with hidden heterogeneous and homogeneous extreme multistability, Extreme multistability in discrete memristive neuron maps and implications for dual-field applications, and Multi-scroll hopfield neural network excited by memristive selfsynapses and its application in image encryption

Response to this suggestion: In section 2, discussion of modern PRNG approaches has been added

4. In Section 3, using the Euler method with a tiny step size introduces correlation between consecutive states. Please reevaluate the discretization strategy or adopt advanced solvers to ensure the generated sequences maintain true unpredictability.

Response to this suggestion: In Section 3.1, provided a more thorough explanation of the Euler method, including its advantages and disadvantages

5. In Section 3, implementing a large constant modular multiplication introduces computational overhead. I suggest optimizing the multiplier block logic and providing a detailed analysis of its practical feasibility and delay costs on the FPGA.

Response to this suggestion: Section 3.4 added to explain the selection for the various parameters

6. In Section 3, the perturbation mechanism modifies chaotic parameters dynamically. Please provide a rigorous mathematical derivation explaining how these specific parameter values guarantee stable global chaotic behavior without collapsing into periodic windows during continuous disruption.

Response to this suggestion: Section 3.4 to further discuss the selection of parameters and perturbations

7. In Section 4, the hardware evaluation omits operating frequency data in the FPGA results. Please report the maximum clock frequency and timing analysis to allow an independent verification of your hardware performance claims.

Response to this suggestion: Section 5.3 added to show the timing analysis

8. In Section 4, the initial key integration relies on static fractional scaling. To prevent linear relationships between key bits and initial states, I recommend introducing a robust, nonlinear key scheduling algorithm to enhance cryptographic security.

Response to this suggestion: In section 6, added the recommendation for future work

9. In Section 5, the resource consumption table lists component counts but excludes latency metrics. Please calculate and present the critical path delay to demonstrate the true hardware efficiency and practical engineering depth of your design.

Response to this suggestion: Section 5.3 added to show the timing analysis

10. In Section 5, the key sensitivity analysis evaluates only a single pair of test keys. Please compute the average Hamming distance across more random key pairs to establish a statistically meaningful cryptographic evaluation

Response to this suggestion: Section 5.4 has been updated to address this concern using 100 keys

11. In Section 5, the uniformity histogram utilizes a relatively small sample size. To

rule out periodic biases and establish long term statistical uniformity, please significantly increase the dataset volume to more generated iterations.

Response to this suggestion: *Section 5.5 has been updated, the dataset has been increased with 25.6 Million iterations*

12. In Section 5, calculating the largest Lyapunov exponent directly from the discrete generator output stream is theoretically problematic. I advise computing this metric exclusively from the continuous phase space trajectory to reflect the true dynamics.

Response to this suggestion: *Section 3.2 has been updated with the Lyapunov exponent of the enhanced system*

13. Please consider updating the reference list to include more recent studies on chaotic systems and their diverse applications. This will help highlight the comprehensiveness, novelty, and superiority of your work.

Response to this suggestion: *Recent studies and research have added to section 2*